

When the cellular traffic changes, how much does an RL scheduler lose?

Measuring the performance loss of AI resource allocation in 5G networks

Mark Zhao & Stanley Findsen Supervisor: Dev Singh University of Auckland

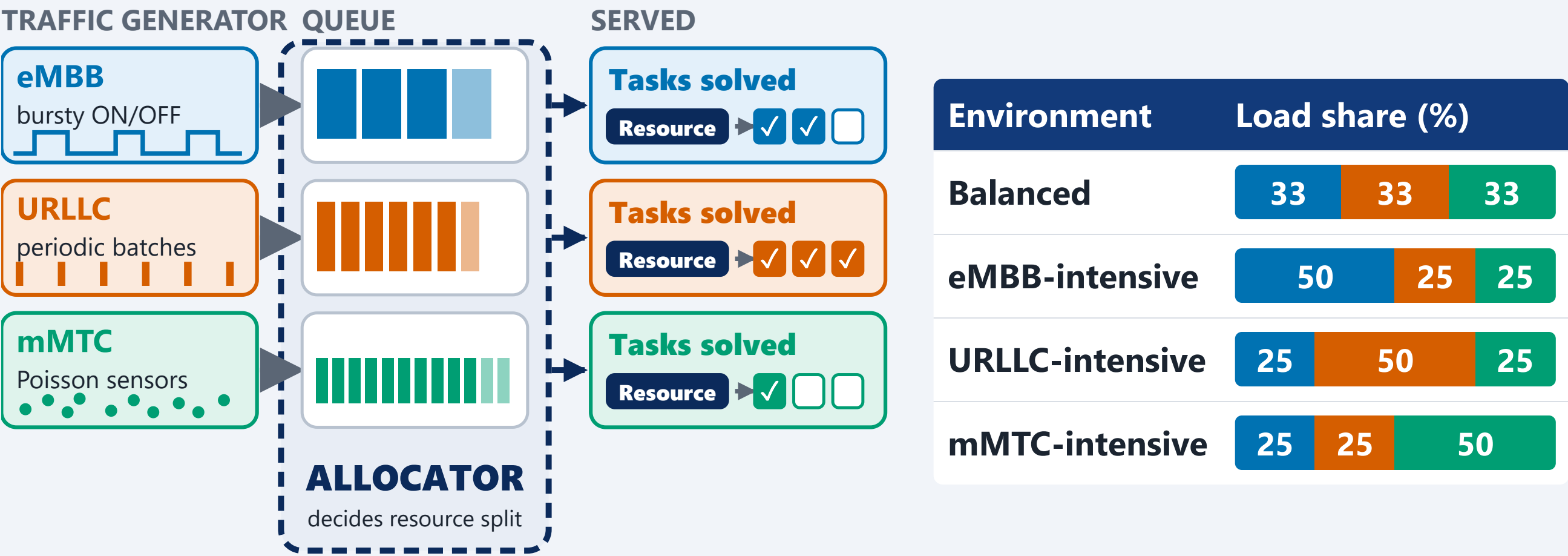
1 The problem

Every millisecond, a 5G base station must share its limited radio capacity between three types of task:

- eMBB** video and downloads
- URLLC** robots and cars
- mMTC** sensor readings

Reinforcement learning (RL) is widely proposed to optimise this decision.

2 Simulator & four environments

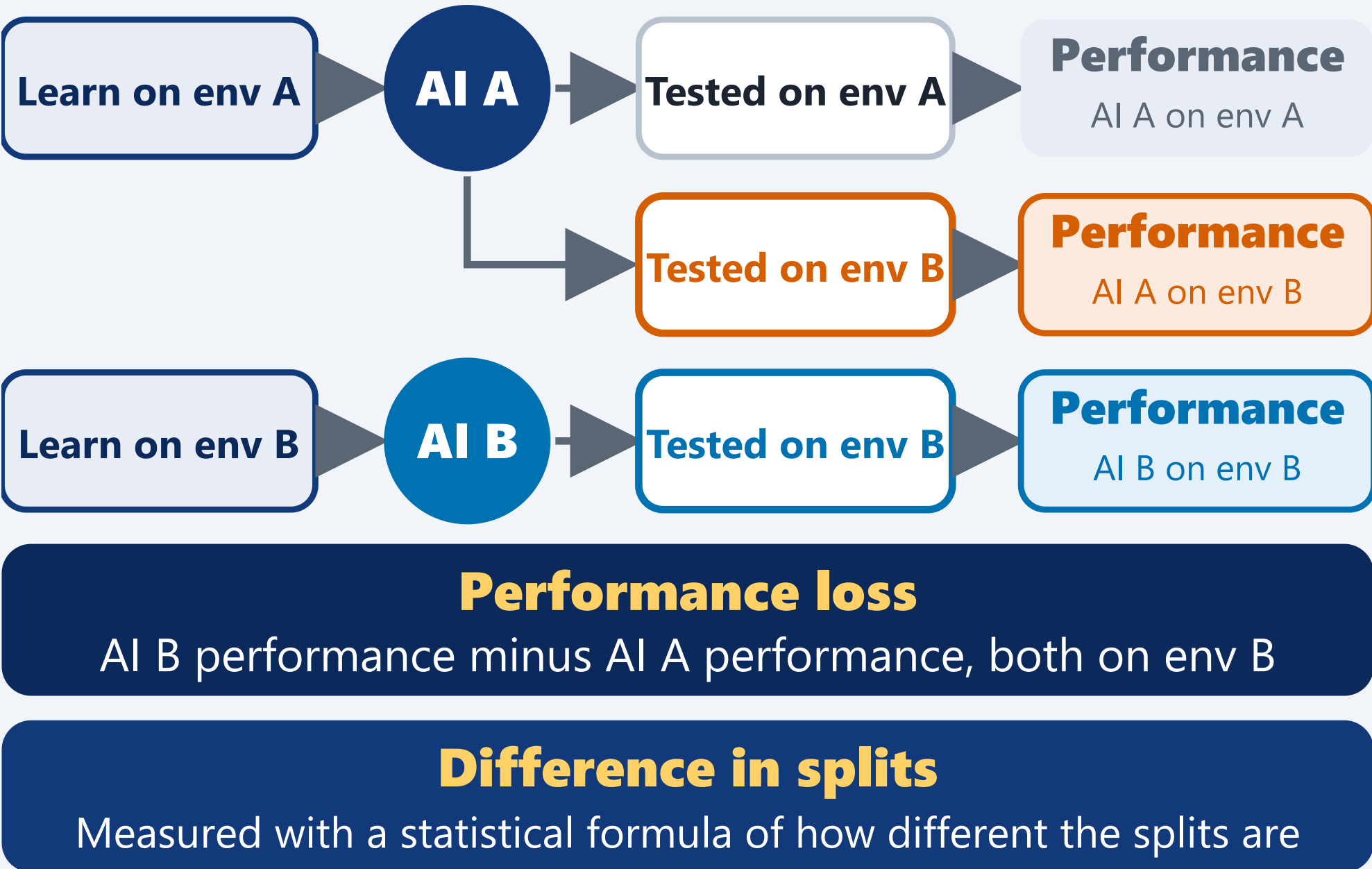


3 How the AI learns

- + REWARD** when tasks in each task type are finished on time.
- PENALTY** when tasks pile up and wait in the queue.

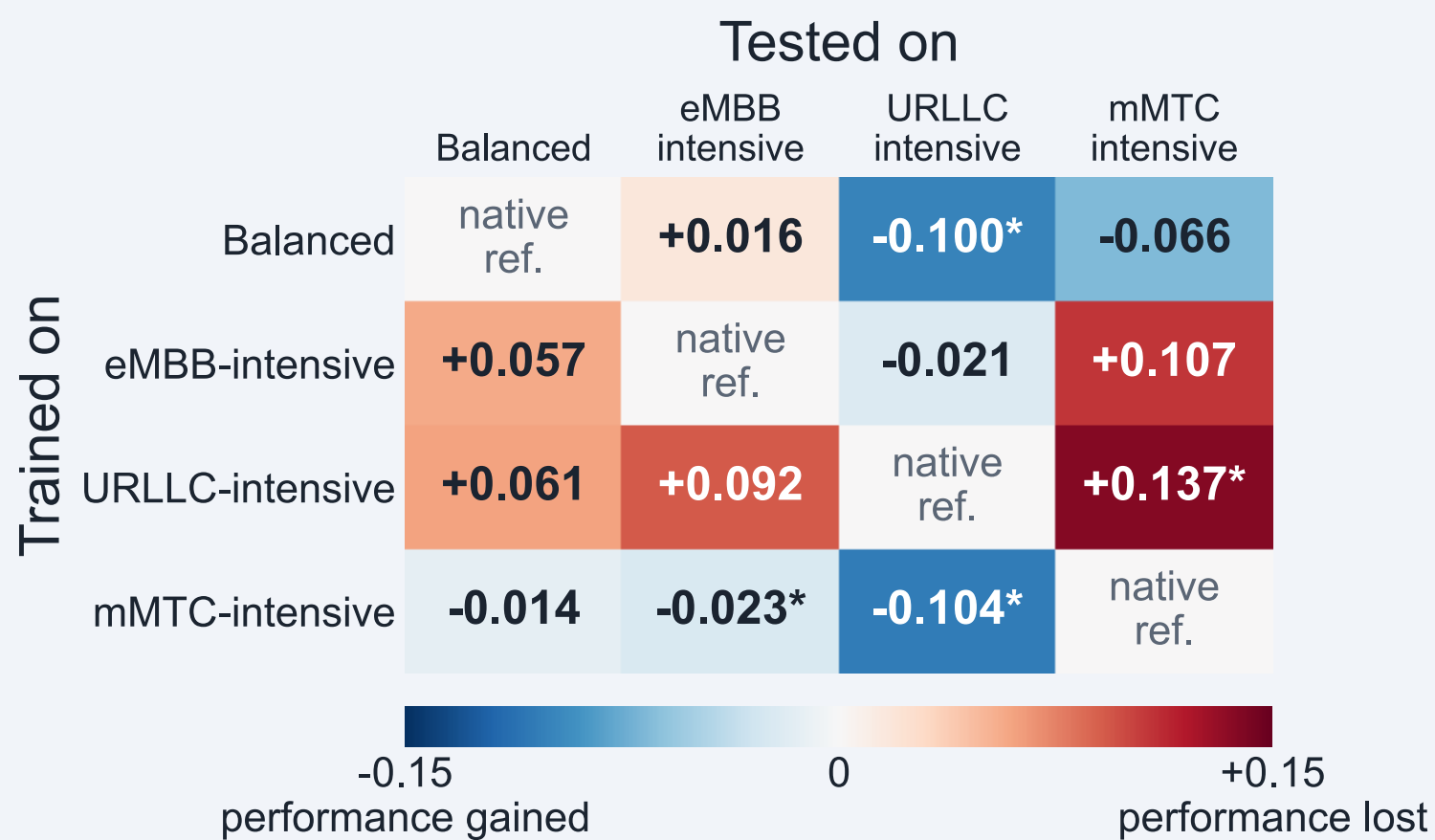
- Learns by** trial and error
- Practice** 5 separate runs per traffic mix
- Length** 600 rounds in each run

4 What has been measured

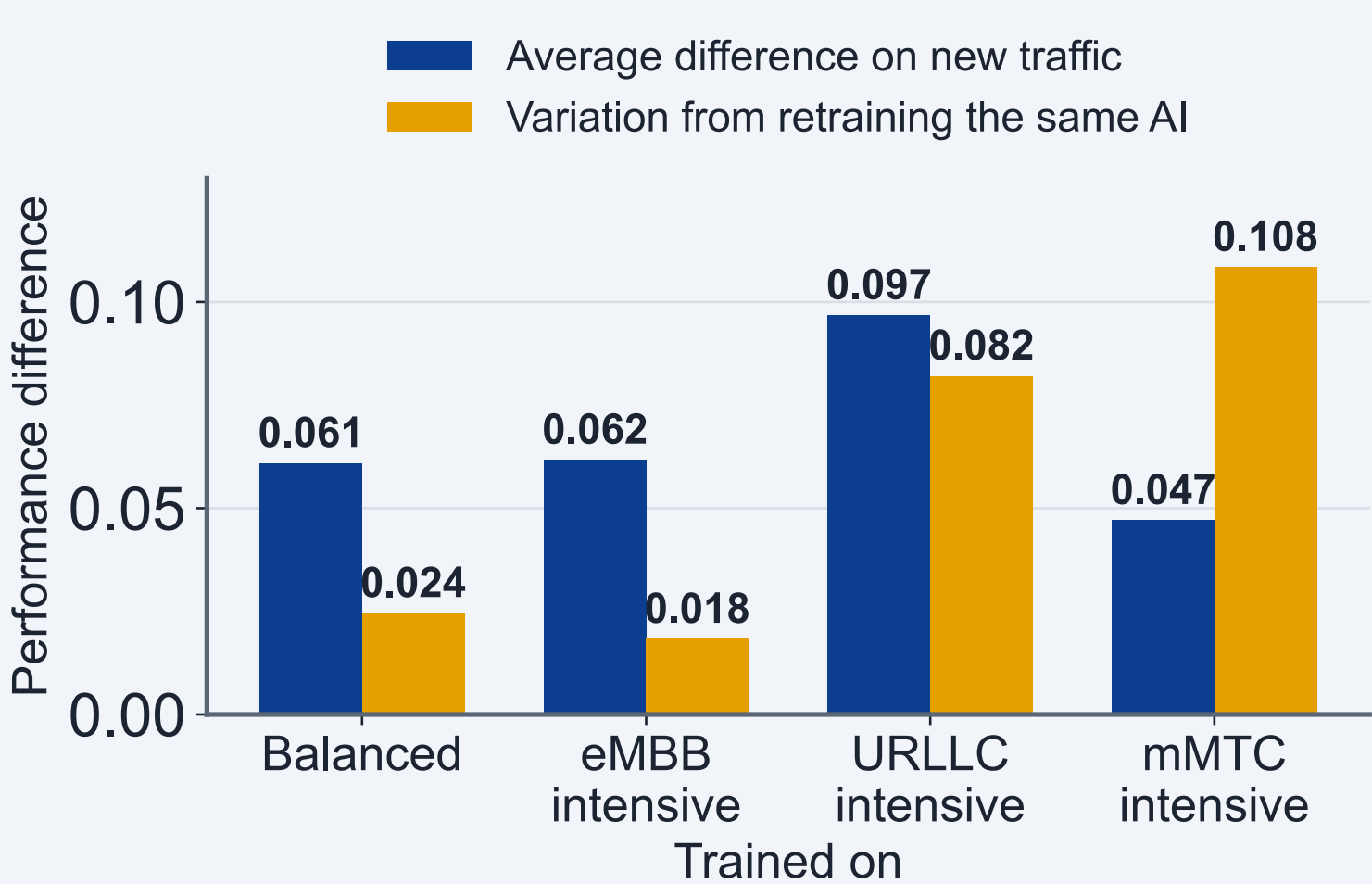


5 Results

Performance loss when an AI is moved to a different traffic type



Performance difference from new traffic, compared with retraining the same AI



6 Conclusions

- An AI trained on **balanced traffic** performs as well as one retrained for each environment, so **there is no need to train a new model** every time the traffic changes.
- Balanced agent transfers well across different traffic conditions.

Contributions

- Mark Zhao**
Built the simulator and RL agent; chose the parameters and assumptions
- Stanley Findsen**
Led the literature review and linked our work to current industry practice